How to Evaluate and Remove the Weakened Bands in Hyperspectral Image Classification

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***Abstract*— Hyperspectral image classification is mainly based on the spectral information of land covers, but water vapor or Rayleigh scattering will weaken the surface reflectance under the effect of adjacent pixels and thus lead to the reduction of the discriminative information for the subsequent classification tasks. Atmospheric correction for the weakened bands is one of the most traditional ways to deal with this issue, but as a complete atmospheric correction for both of them is difficult, maybe a systematic exclusion of the severely affected bands based on quantitative evaluation is a better choice. In this article, an evaluation-based weaken band exclusion method for the hyperspectral image classification is proposed, trying to remove the severely affected bands without further atmospheric correction. Specifically, an evaluation model to describe how the water vapor and Rayleigh scattering affect the surface reflectance is constructed, by using the statistical relationship between the radiative transfer model and the band weaken index of spectra among the adjacent pixels. Then, with a simulation experiment, it is shown that water vapor and Rayleigh scattering can really weaken the discriminative information of some specific bands, and the band weaken index can serve as an appropriate index to evaluate the weakening degree of those bands. Finally, on this basis, the total framework of evaluation-based weaken band exclusion method is given. The effectiveness and the universality of our proposed method have been verified and compared on four representative tasks of the hyperspectral image classification.**

***Index Terms*— Band weaken index, evaluation model, hyperspectral image (HSI), land cover classification, weaken band exclusion.**

I. Introduction

HYPERSPECTRAL image (HSI) consists of hundreds of spectral bands and can reflect the inherent properties and physical meanings of land covers. Benefiting from these, HSI has been widely used in various application fields, such as land cover classification [1], [2], precise identification [3], [4], vegetation estimation [5], environmental monitoring [6], and even for the landslide detection [7].

Received 10 October 2024; revised 14 December 2024; accepted 3 January 2025. Date of publication 8 January 2025; date of current version 17 January 2025. This work was supported in part by the National Nature Science Foundation under Grant 82471999 and Grant 41971294, in part by Beijing Natural Science Foundation under Grant 3254044, and in part by Beijing Institute of Technology Research Fund Program for Young Scholars of China. *(Corresponding authors: Xiaolin Han; Weidong Sun.)*

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Digital Object Identifier 10.1109/TGRS.2025.3526917

Generally speaking, HSI classification is based on the spectral information of land covers [8], [9], but water vapor or Rayleigh scattering may weaken or even flatten the surface reflectance under the effect of adjacent pixels and thus lead to the reduction of the discriminative information for subsequent classification tasks. To deal with the weakened or unstable bands, atmospheric correction should be conducted first, but a complete atmospheric correction is theoretically impossible, and the influence of water vapor or Rayleigh scattering may still exist even after the atmospheric correction. Therefore, maybe shielding or removing the most affected bands based on statistic distribution in local regions instead of a complete atmospheric correction is another possible solution [10], which can be classified as a specialized technique of band selection [11].

For band selection, it can be described with two opposing views: one removes the problematic bands, while the other retains the appropriate bands [10], [12]. Nearly, all of the band removing methods focus on the removal of the noise bands or low SNR bands [13], and the band retaining methods mainly focus on the selection of the most representative bands for some specific application needs [14], [15], [16], [17].

Moreover, the strategies of band selection for HSI can be divided into five major categories: projection-based [18], [19], [20], [21], [22], [23], ranking-based [24], [25], [26], [27], [28], clustering-based [29], [30], [31], [32], [33], [34], [35], deep learning-based [37], [38], [39], and hybrid-based band selection [15], [16], [17], [40].

Projection-based band selection usually takes the statistical characteristics between spectral bands into consideration and accomplishes the selection through feature compression, in which the feature dimension of HSIs will be contracted, to condense numerous weak features into fewer strong features, by using the traditional methods such as principal component analysis (PCA) [18], independent component analysis (ICA) [19], linear discriminant analysis (LDA) [20], and partial least-squares discriminant analysis (PLS-DA) [21]. Among them, Li et al. [22] utilized the local Fisher's discriminant analysis (LFDA) to reduce the dimensionality of features while reserving structures. Li et al. [23] introduced locality preserving projection (LPP), to reduce features and reserve the local property. In these methods, by assessing the validity of features, the bands corresponding to the features with strong discrimination ability are selected and reserved. However, the physical meaning of HSI is generally destroyed during the projection [2].

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When it comes to the ranking-based band selection, these methods adopt different evaluation criteria to estimate the significance of each band and select the top-ranked ones, by using evaluation criteria such as information entropy [24], information divergence [25], and variance inflation factor (VIF) [26]. Among them, Morales et al. [27] proposed a pre-selection method to filter the subset of independent bands, followed by an information entropy-based greedy selection. Wang et al. [28] divided adjacent bands into connected groups and then employed the regional informative mutuality ranking on another larger pixel patch within each homogeneous band group, to acquire a band subset. Even though these methods have validated their effectiveness, they only take HSI as a multidimensional signal and ignore the internal relationship between the spectral bands and the physical imaging models, especially the internal relationship between surface reflectance, imaging time, and the categories of land cover.

In recent years, clustering-based band selection has gradually emerged, which first partitioned the bands into several subspaces by clustering [29], [30] and then selected the optimal band from each subspace to form a new subset [31]. Wang et al. [32] proposed an optimal clustering framework (OCF), which searched for an optimal band partition by using the clustering strategy, followed by a ranking and selection. On this basis, they also proposed a fast neighborhood grouping method for hyperspectral band selection (FNGBS), which partitioned the sample data into groups by assessing similarity and then obtained the most informative bands by using local density and information entropy [33]. In addition, Xu et al. [34] utilized Structural SIMilarity (SSIM) to evaluate each band being a cluster center, among which the top-ranked bands are selected. Sun et al. [35] designed an adaptive distance-based band hierarchy clustering to acquire a flexible number of band subset by leveraging a tree hierarchy-based data structure of HSI. Ma and Yang [36] proposed a spatial–spectral hypergraph-based unsupervised band selection (SSHUBS) method, in which both spatial hypergraph and spectral hypergraph were built to embed spatial and spectral information into the HSI band selection. We can see that although a band subset can be obtained from the clusters, the clustering methods themselves may mislead the results of the subsequent classification task, due to the difference between unlabeled data grouping and data categorizing with known labels, especially some of the most informative and discriminative bands may be merged into large categories for their smaller presence [14].

With the development of deep learning [37], deep learning-based band selection appeared, which focused on learning the optimal combination of spectral band. Among them, Taherkhani et al. [38] regularized the convolutional filters of the first layer of a convolutional neural network (CNN) by using a group of least absolute shrinkage and selection operators (LASSOs), to sparsity the redundant bands. Tang et al. [10] introduced a band selection method by using spectral clustering via spatial–spectral weighted and region-wise multiple graph fusion. Similar attempts have also been carried out in [39], where a spectral-wise attention mechanism in the fully connected layer is applied to emphasize informative spectral features. However, one common problem with the aforementioned band selection methods is that they generally ignore the influence of the atmosphere on the spectral bands.

For the last category, hybrid-based band selection has also been investigated, which tries to integrate two or more methods for the band selection, such as intraclass separability index (SSI) [15] and manifold ranking [16]. Among them, Walton et al. [40] proposed a method named histogram-assisted genetic algorithm for reduction in dimensionality (HAGRID), which maintains a population of index vectors identifying a specific number of wavelengths and fits a Gaussian mixture model to identify the main wavelengths with their corresponding filter bandwidths. Wang et al. [17] integrated ranking and network pruning to retain a compact subnetwork for the band selection and HSI classification. Even though they integrated different strategies, in the practical applications, when the imaging condition with water vapor and Rayleigh scattering changes, the discriminative information of spectral bands will also change accordingly, so the existing band selection methods will easily fail. In addition, the weakened discriminative information of spectral bands not only affects the HSI classification but also affects the spectral unmixing [41], [42], [43].

To solve the above problems, an evaluation-based weaken band exclusion method for the hyperspectral image classification, termed WBE, is proposed in this article, trying to remove the severely affected bands without further atmospheric correction. Specifically, an evaluation model to describe how the water vapor and Rayleigh scattering affect the surface reflectance is constructed, by using the statistical relationship between the radiative transfer model and the band weaken index of spectra among the adjacent pixels, i.e., the proportional relationship between the atmospheric point spread function and the band weaken index. Then, with the results of a simulation experiment, it is shown that water vapor and Rayleigh scattering can really weaken the discriminative information of specific bands, and the band weaken index can serve as an appropriate index to evaluate the weakening degree of those bands. Finally, on this basis, the total framework of the evaluation-based weaken band exclusion method is given. Also, the effectiveness and the universality of our proposed method have been verified and compared on four representative tasks of the hyperspectral image classification. The novelty and contributions of our proposed method can be summarized as follows.

1) A new evaluation model to describe how the water vapor and Rayleigh scattering affect the surface reflectance is constructed, by using the statistical relationship between the radiative transfer model and the band weaken index of spectra among the adjacent pixels.
2) A well-designed simulation has been carried out to show that water vapor and Rayleigh scattering can really weaken the discriminative information of some specific bands, and the novel band weaken index can serve as an index to evaluate the weakening degree of them.
3) An evaluation-based weaken band exclusion method for the hyperspectral image classification is proposed,

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TABLE I
IMPORTANT SYMBOLS

L^*	Radiance of each pixel at the sensor
ρ	Surface reflectance of the pixel
ρ_e	Average surface reflectance of the surrounding pixels
$< \rho >$	Ambient reflectance
$\mathbf{\Psi}_{PSF}$	Atmospheric Point Spread Function (PSF)
$PSD_\Psi(f_x, f_y)$	Power spectral density of the PSF
$PSD_I(f_x, f_y)$	Power spectral density of the observed image
PSD_T	Power spectral density of transmission function
$\mathbf{\Upsilon}(\Delta x, \Delta y)$	Spatial correlation function
γ	Band weaken index

to remove the affected bands by water vapor or Rayleigh scattering. To the best of our knowledge, this is the first band selection method that can improve the accuracy of HSI classification.

The remainder of this article is organized as follows. Our proposed method is introduced in Section II. Section III presents the experimental results on band selection and four different tasks of the hyperspectral image classification, and a conclusion is given in Section IV.

II. PROPOSED METHOD

In this section, as shown in Fig. 1, the problem formulation concerning with our proposed method will be illustrated first. Then, an evaluation model to describe how the water vapor and Rayleigh scattering affect the surface reflectance will be given. After that, simulation experiments will be conducted, to validate that water vapor and Rayleigh scattering can really weaken the discriminative information, and the band weaken index can serve as an appropriate index to evaluate the weakening degree. Finally, the total framework of our proposed method will be introduced, among which the important symbols are listed in Table I.

A. Problem Formulation

The hundreds of spectral bands of HSI can reflect the inherent properties and physical meanings of land covers, but if we use it as a space-borne or airborne sensor, the effects of atmosphere need to be considered first. Taking the NASA AVIRIS spectrometer as an example, the radiance L^* of each individual pixel received at the sensor can generally be formulated as follows [44]:

$$L^* = \frac{A\rho}{1 - \rho_e S} + \frac{B\rho_e}{1 - \rho_e S} + L^*_\alpha \tag{1}$$

where A and B are determined by the atmospheric and geometric conditions over the observation surface, respectively; ρ is the surface reflectance of the pixel; ρ_e is the average surface reflectance of the surrounding pixels; S denotes the atmospheric spherical albedo; and L^*_α denotes the atmospheric path radiation. To simplify this formula, the wavelength index in (1) is omitted as in [45]. The main difference between ρ and ρ_e comes from the effect of adjacent pixels. However, most of the atmospheric correction models or the well-known software packages, which extract spectral information to convert data from radiance to reflectance, such as Atmospheric CORrection Now (ACORN) [46], High-accuracy ATmospheric Correction for Hyper-spectral data (HATCH) [47], and ATmospheric REMoval program (ATREM) [48], have ignored this kind of effect and generally set $\rho = \rho_e$. This kind of simplicity is feasible under the conditions of high visibility and uniform surface, but it will lead to significant errors under the conditions of complex atmospheric condition or nonuniform surface.

Specifically, ATREM software converts the AVIRIS data to surface reflectance, modeled after the MODerate resolution atmospheric TRANsmission (MODTRAN) and 5S radiative transfer code [49], in which the surface is assumed to be a horizontal surface with a Lambertian reflectance. Also, the atmospheric correction results for all spectral bands at a given pixel in the Kennedy Space Center (KSC) dataset are shown in Fig. 2 [52], where the values were multiplied by log10 to be plotted on the same scale. Fig. 2 indicates that even if ATREM can correct the atmospheric effect in visible bands, it overcompensated the correction for the bands in water absorption regions, especially for bands 102–116 and bands 150–172, where as shown in the two black dotted boxes, the curve with red exceeds that with purple. In other words, the correction results outside the visible bands may be inaccurate.

Moreover, the spectra of each category in this KSC dataset after removing the two regions affected mostly by the water absorption, i.e., bands 102–116 and bands 150–172 shown with black dotted boxes, are also given in Fig. 3. We can also see with Fig. 3 that even after the removing of water absorption regions, some abnormal intensity maxima still exist in several bands near the regions of bands 150 and 180–224 shown with the light-blue dotted boxes, which indicates that the removing of water absorption is still inadequate for this dataset. Furthermore, combined with Fig. 2, we can also find that, in the regions shown with light-blue boxes, the curve with the red color is also slightly exceeded that with the purple color, which has a strong correspondence with the light-blue boxes in Fig. 3 coincidentally.

The above results indicate that when the atmospheric condition is complex or the surface is not approximately a uniform Lambertian surface, ignoring the effect of the adjacent pixels just by simply set $\rho = \rho_e$ would lead to unreliable atmospheric correction results. Even after removing bands affected mostly by the water vapor absorption, there are still some intensity maxima existing in the spectra that could still affect the subsequent classification tasks.

B. Evaluation Model for the Discriminative Information

To better describe what led to those significant differences, the general surface reflectance at each individual pixel of the sensor is given as follows [53]:

$$\rho^* = \frac{\left(e^{-\frac{\tau}{\mu_s}} + t_d(\theta_s)\right)\left(e^{-\frac{\tau}{\mu_v}}\rho + t_d(\theta_v)\rho_e\right)}{1 - \rho_e S} + \rho_\alpha \tag{2}$$

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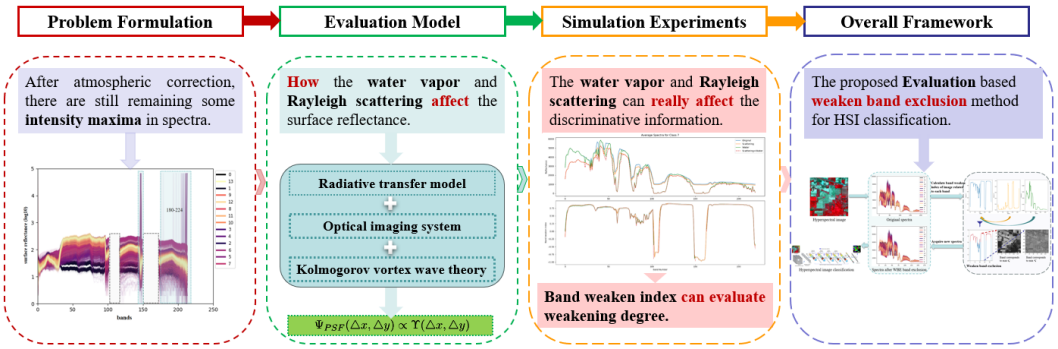


Fig. 1. Overall workflow of the Section II.

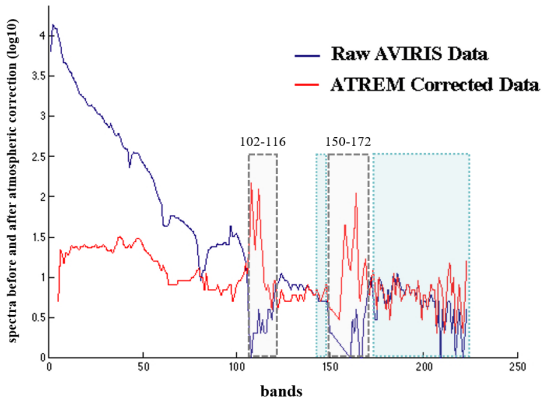


Fig. 2. Comparison of raw AVIRIS data and the atmospheric correction results from ATREM for a given pixel.

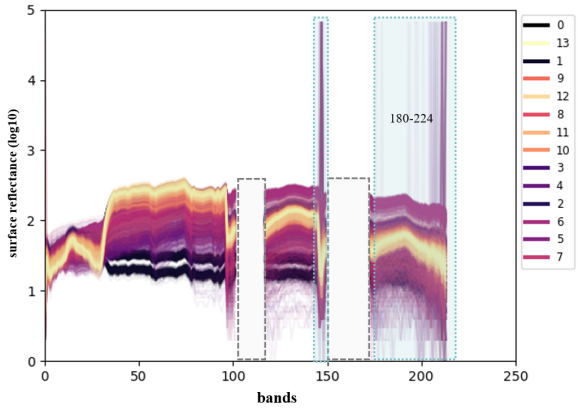


Fig. 3. Surface reflectance of each category in the KSC dataset, where the legend indicates the sorting of categories from large to small.

where S, ρ, and ρ_e are consistent with that in (1); ρ_α denotes the atmospheric reflectance; $e^{-(\tau/\mu_s)} + t_d(\theta_s)$ represents the sum of direct transmittance and scattering transmittance in the incident direction of the sun; $e^{-(\tau/\mu_v)}$ represents the direct transmittance in the observation direction; $t_d(\theta_v)$ represents the scattering transmittance in the observation direction; and $t_d(\theta_v)\rho_e$ represents the radiation reflected by the surrounding pixels outside the field of view. If the reflectance of the surrounding pixels is nonuniform and with complex structures, an ambient reflectance $\langle\rho\rangle$ is generally defined, to represent the spatial average of the surface reflectance of all pixels on the observed surface [45]

$$\langle\rho\rangle = \frac{1}{t_d(\theta_v)} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \rho(x, y) e(x, y, \theta_v) dx dy. \quad (3)$$

Currently, the origin is $(0, 0)$, $\rho(x, y)$ denotes the surface reflectance of the pixel $p(x, y)$, and $e(x, y, \theta_v)$ represents the contribution of each pixel to the scattering transmittance $t_d(\theta_v)$. When the distances between the surrounding pixels and the target pixel exceed a certain range, the contribution of surrounding pixels to the target pixel is quite fractional, which can be ignored. Thus, (3) can be rewritten as

$$\langle\rho\rangle = \sum\sum \rho(\Delta x, \Delta y) \mathbf{\Psi}_{\text{PSF}}(\Delta x, \Delta y, \theta_v) \quad (4)$$

where $\mathbf{\Psi}_{\text{PSF}}(\Delta x, \Delta y, \theta_v)$ is a weight function, which describes the contribution of the pixel with a distance of $(\Delta x, \Delta y)$ to the target pixel. Furthermore, $\mathbf{\Psi}_{\text{PSF}}$ is independent of the characteristics of surface, which is known as the atmospheric point spread function (PSF). Also, the PSF reflects the contribution of surrounding pixels to the target pixel under the effect of adjacent pixels [45].

As mentioned in [54], the main influencing factors to the PSF are atmospheric turbulence, water vapor, and Rayleigh scattering. Among them, atmospheric turbulence has the greatest impact under extreme weather conditions, severe terrain fluctuations, and high-resolution observations. Also, water vapor and Rayleigh scattering have the most extensive impact on the PSF, because of their ubiquitous characteristics.

Atmospheric turbulence can hardly be corrected by atmospheric correction models, among which the water vapor and the Rayleigh scattering are the dominant factors [69]. In fact, unstable atmospheric turbulence is usually presented as a sort of spatial distribution pattern with intensity changes across time, which is difficult to accurately capture, as illustrated in [70]. Therefore, the imaging process of ground surface through unstable atmospheric turbulence can be modeled as a discrete set of phase screens by the split-step propagation [71] as shown in Fig. 4, where the wavefront distortion is caused by cropping regions of the phase screen at each pixel. The key operations within this process are Fresnel propagation and Kolmogorov phase imparting. The final result of a sequence of these operations is a PSF for each pixel. Moreover, the overlaps of the phase screens create a spatial correlation [72].

Furthermore, the unstable atmospheric turbulence will affect land covers severely, by changing local optical parameters

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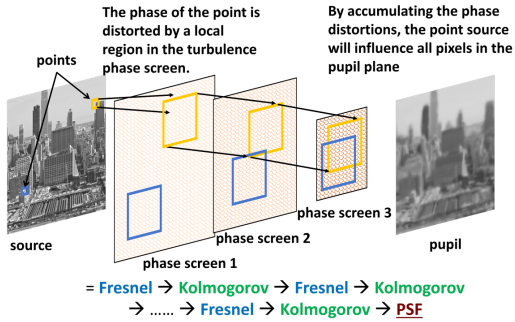


Fig. 4. Overlaps of the phase screens create a spatial correlation on different phases [72].

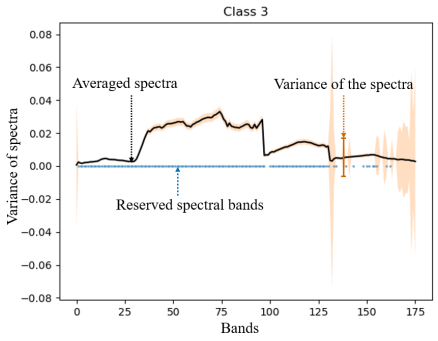


Fig. 5. Atmospheric turbulence makes the surface reflectance unstable, which can be observed by the variance of spectra.

and disrupting the deterministic relationship between adjacent pixels. Finally, the surface reflectance in local regions will be unstable, which weakens the imaging process of corresponding spectral bands and leads to the decrease of spatial correlation due to the split-step propagation. This kind of unstable surface reflectance can be observed by the variance of spectra, as shown in Fig. 5. From Fig. 5, it can be seen that the large variance, i.e., the unstable surface reflectance brought by the unstable atmospheric turbulence, can be filtered out by the quantitative evaluation index of spatial correlation. Therefore, it is clear that the quantitative evaluation index of spatial correlation maybe can serve as an appropriate index to evaluate the weakening degree of spectral bands.

In terms of conceptual expression, for a given HSI, θ_v is constant, which can be omitted in $\mathbf{\Psi}_{\mathrm{PSF}}(\Delta x, \Delta y, \theta_v)$, while PSF corresponds to the different wavelengths of each band. Therefore, for the band k, according to the theory of Fourier analysis [55], there is the 2-D inverse Fourier transform relationship between the power spectral density of the PSF (PSD_Ψ) and the atmospheric point spread function $\mathbf{\Psi}_{\mathrm{PSF}}$

$$\mathbf{\Psi}_{\mathrm{PSF}}(\Delta x, \Delta y) = \mathrm{PSD}_\Psi\left(f_x, f_y\right)e^{j2\pi\left(f_x\Delta x + f_y\Delta y\right)}df_x df_y \quad (5)$$

where (f_x, f_y) denotes the 2-D frequency coordinate. Since the power spectral density of the PSF $\mathrm{PSD}_\Psi(f_x, f_y)$ cannot be directly measured or calculated by frequency response, it is necessary to utilize observed image data and transmission models for the calculation. Concretely, according to the model of optical imaging system [55], the $\mathrm{PSD}_\Psi(f_x, f_y)$ can be calculated by the power spectral density of the observed image $\mathrm{PSD}_I(f_x, f_y)$ that corresponds to the band k and the power spectral density of the transmission function PSD_T

$$\mathrm{PSD}_\Psi\left(f_x, f_y\right) = \mathrm{PSD}_I\left(f_x, f_y\right)/\mathrm{PSD}_T \quad (6)$$

among which, the power spectral density of the transmission function PSD_T can be calculated through the Kolmogorov vortex wave theory model [56], which describes the influence of the atmosphere on the transmission function of optical imaging systems and is widely used to calculate and analyze the performance of atmospheric optical imaging systems

$$H(f) = e^{-3.44\left(\frac{\lambda R f}{r_0}\right)^{\frac{5}{3}}} \quad (7)$$

where λ denotes the wavelength of band k; R denotes the radius of wavefront; f denotes the 1-D frequency and it can be calculated by $f = (f_x^2 + f_y^2)^{1/2}$; and r_0 denotes the magnitude of the atmospheric vortex wave, which is a constant. After obtaining the frequency response of the transmission function, $H(f)$, the power spectral density of the transmission function PSD_T can be expressed as

$$\mathrm{PSD}_T = |H(f)|^2. \quad (8)$$

On the other hand, the other item in (6), i.e., $\mathrm{PSD}_I(f_x, f_y)$, gives an observed image that corresponds to the specific band, and its related spatial correlation function $\mathbf{\Upsilon}(\Delta x, \Delta y)$ can be acquired by conducting the 2-D inverse Fourier transform [57] on the power spectral density of the image $\mathrm{PSD}_I(f_x, f_y)$ [58]

$$\mathbf{\Upsilon}(\Delta x, \Delta y) = \mathrm{PSD}_I\left(f_x, f_y\right)e^{j2\pi\left(f_x\Delta x + f_y\Delta y\right)}df_x df_y. \quad (9)$$

Since there is no $(\Delta x, \Delta y)$ item in PSD_T, combined with (5) and (9), the formulation for each band k can be denoted as follows:

$$\mathbf{\Psi}_{\mathrm{PSF}}(\Delta x, \Delta y) \propto \mathbf{\Upsilon}(\mathbf{\Delta} x, \Delta y). \quad (10)$$

From the above proportional relationship between the PSF $\mathbf{\Psi}_{\mathrm{PSF}}$ and the spatial correlation function $\mathbf{\Upsilon}$, which is a new theoretical contribution, we can see that $\mathbf{\Upsilon}$ can really be used to evaluate the effects of atmospheric condition, and thus, the weakening degree of the discriminative information affected by the water vapor and Rayleigh scattering. As a result, if the spatial correlation of a specific band is much low, we can regard that it has been seriously affected by the water vapor and Rayleigh scattering, and the discriminative information within it is not significant or even distorted; then, this band should be excluded in the following classification task.

C. Verification of the Evaluation Model

To demonstrate that the spatial correlation can be utilized to evaluate the weakening degree of the discriminative information in each band, the most representative atmospheric correction model 6S [59] is applied, to simulate the influence of water vapor and Rayleigh scattering in the hyperspectral images. Specifically, 6S calculates the water vapor and Rayleigh scattering of each band based on their wavelength and then multiplies them with a corresponding grayscale image. Here, the Indian Pines (IP) hyperspectral image dataset [50] is selected for the simulation experiments. Generally, for an individual image related to a specific band k,

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the spatial correlation function can be calculated by the value of each pixel $\rho(x, y)$ and that of its surrounding pixel $\rho(x', y')$

$$\Upsilon(\Delta x, \Delta y) = \mathbb{E}\big[\rho(x, y) - \bar{\rho}\big]\big[\rho\big(x', y'\big) - \bar{\rho}\big] \tag{11}$$

where $\Delta x = x - x'$, $\Delta y = y - y'$, $\mathbb{E}$ denotes expectation, and $\bar{\rho}$ denotes the average value. Here, for the calculation of the above spatial correlation function with normal distribution, a band weaken index accounting for the surrounding pixels $p(x, y)$ and variance of spectra σ is proposed as follows:

$$\gamma_I = \frac{n}{\Omega_I} \times \frac{\sum\sum \omega_{\Delta x,\Delta y}\big(\rho(x, y) - \bar{\rho}\big)\big(\rho\big(x', y'\big) - \bar{\rho}\big)}{\sum\big(\rho(x, y) - \bar{\rho}\big)^2} \tag{12}$$

$$\gamma_C = \frac{n-1}{2\Omega_C} \times \frac{\sum\sum \omega_{\Delta x,\Delta y}\big(\rho(x, y) - \big(\rho\big(x', y'\big)\big)\big)^2}{\sum\big(\rho(x, y) - \bar{\rho}\big)^2} \tag{13}$$

$$\gamma = (1 - \sigma) \times (\gamma_I - \gamma_C) \tag{14}$$

where n is the amount of pixels in the area; $\omega_{\Delta x,\Delta y}$ denotes the weight corresponding to the proximity relationship of the two pixels; and Ω represents the set of the weights, i.e., $\Omega_I = \{\omega_{\Delta x,\Delta y}\}$ and $\Omega_C = \{\omega_{\Delta x,\Delta y}\}$. σ denotes the variance of spectra; γ_I and γ_C denote the Moran index [60] and the Geary index [61], respectively; and the final band weaken index $\gamma \in [0, 1]$.

Simulation experiment results are shown in Fig. 6, and the averaged spectrum of a category in the IP dataset is selected to analyze the influence of water vapor and Rayleigh scattering, in which blue curves represent the original averaged spectrum, yellow curves represent the averaged spectrum under the influence of Rayleigh scattering, green curves represent the averaged spectrum under the influence of water vapor, and the red dotted lines represent the averaged spectrum under the superposition of these two impacts.

It can be seen from Fig. 6(a) that when the averaged spectrum is affected by the water vapor and Rayleigh scattering, their reflectance value will be weakened as shown in the red boxes, and the discriminative information will be reduced. What is more, as shown in Fig. 6(b), the band weaken index can correspond well to the weakening degree of the spectrum affected by water vapor and Rayleigh scattering, i.e., the smaller the variance and the bigger the Moran index minus the Geary index are, the larger the band weaken index will be, and those bands with smaller band weaken index should be excluded. Then, the band weaken index can serve as an appropriate index to evaluate the weakening degree of discriminative information of a specific band.

To quantify the ability of the band weaken index, both the weakening degree of discriminative information and the changing of band weaken index are binarized as follows:

$$\phi_k = \begin{cases} \zeta_0, & \xi_k < \beta_1(\xi_{\max} - \xi_{\min}) \\ \zeta_1, & \xi_k \geq \beta_1(\xi_{\max} - \xi_{\min}) \end{cases} \tag{15}$$

$$\gamma_k = \begin{cases} \zeta_0, & \gamma_k < \beta_2(\gamma_{\max} - \gamma_{\min}) \\ \zeta_1, & \gamma_k \geq \beta_2(\gamma_{\max} - \gamma_{\min}) \end{cases} \tag{16}$$

where ϕ_k denotes the label of the reflectance of the kth band, ζ denotes the binarized label, and $\xi_k \in (\xi_{\min}, \xi_{\max})$ denotes the weakening degree of the discriminative information of each band. $\gamma_k \in (\gamma_{\min}, \gamma_{\max})$ denotes the values of band weaken index of the kth band. The threshold of weakening degree of discriminative information β_1 is approximate to the threshold of band weaken index β_2, i.e., $|\beta_1 - \beta_2| < 0.2$. Then, the weakening degree of discriminative information can be given in Fig. 6. On this basis, by calculating the following formulation:

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{17}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{18}$$

$$F\text{-score} = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}. \tag{19}$$

TABLE II
QUANTITATIVE RELATIONSHIP ON THE IP DATASET

Band weaken index	Precision	Recall	F-score
γ	80.67	79.36	79.57

A quantitative evaluation for the band weaken index on the IP dataset has been obtained, as shown in Table II.

From Table II, it can be clearly seen that the band weaken index does have a stable credibility to serve as an appropriate index to evaluate the weakening degree of each spectral band, which is a novel theoretical contribution.

D. Band Weaken Index-Based Band Exclusion Method

After verifying that the band weaken index can be used to evaluate the weakening degree of discriminative information, our newly designed evaluation-based weaken band exclusion method for hyperspectral image classification, the WBE, will be given in this section, to remove the weakened bands affected by the water vapor and Rayleigh scattering. Specifically, for a given band to be evaluated, the band fixed HSI will be regarded as a grayscale image and its band weaken index will be calculated first as in (14), and then, it will be conducted to remove the bands whose band weaken index is lower, and the bands whose band weaken index is greater than a given threshold will be reserved.

Then, band weaken index can be filtered by adjusting the threshold value of β_2, some specific bands can be selected according to the value of band weaken index. Also, the overall framework of our above proposed weaken band exclusion method[1] is given in Fig. 7 and Algorithm 1.

III. EXPERIMENTAL RESULTS AND DISCUSSION

In this section, the band selection results of our proposed method will be first given. Then, the effectiveness and the universality of our proposed method will be verified on four representative tasks of the hyperspectral image classification.

[1]https://github.com/Qingyanyichen/WBE-method

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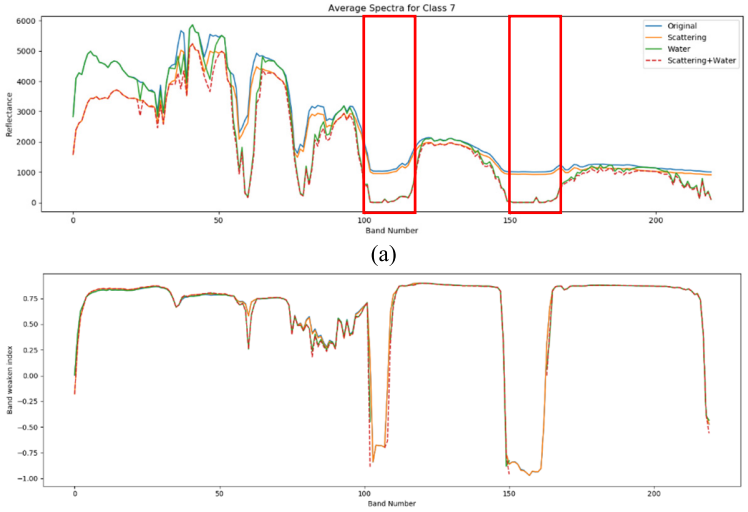


Fig. 6. Relationship between (a) averaged spectrum of a category in the IP dataset under the influence of water vapor and scattering, and (b) corresponding band weaken index. (a) Averaged spectrum of a category in the IP dataset under the influence of water vapor and scattering. (b) Corresponding band weaken index.

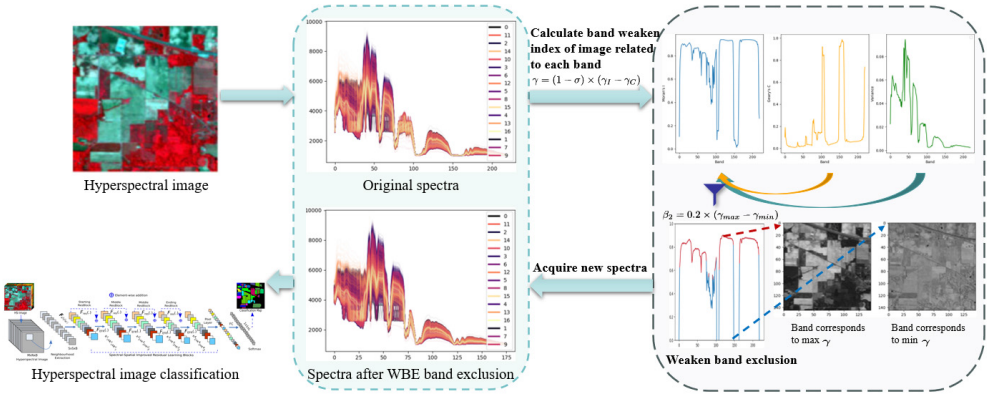


Fig. 7. Overall framework of our proposed method.

Algorithm 1 Weaken Band Exclusion Method

Input: HSI data $\boldsymbol{P}_{ori} \in \mathbb{R}^{H \times W \times N}$;
Parameters setting: β_2, threshold of band weaken index;
Output: HSI data after band selection $\boldsymbol{P} \in \mathbb{R}^{H \times W \times N'}$

For i = 1 to N **do**
  Calculate $\gamma_{i,I}$ as in Eq. (12)
  Calculate $\gamma_{i,C}$ as in Eq. (13)
  Calculate γ_i as in Eq. (14)
End
Weaken band exclusion according to Eq. (16)

A. Datasets in the Experiments

Four representative and influential HSI classification datasets, including IP, Salinas, Xiongan New Area, and KSC datasets, are utilized to evaluate our proposed WBE method.

The IP dataset was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) over the IP in Northwestern Indiana, with a moderate spatial resolution of 20 m [62]. It consists of 145 × 145 pixels, among which 10 249 pixels are labeled and divided into 16 categories. In this dataset, 220 spectral bands in the wavelength range of 400–2500 nm are provided for public testing.

The Salinas (SA) dataset was collected by the AVIRIS over the Salinas Valley, California, and is characterized by high spatial resolution [62]. This dataset comprises 224 spectral bands, 512 × 217 pixels, and is divided into 16 categories of land covers.

The aerial hyperspectral classification dataset of Xiongan New Area (XA) is collected by a full-spectral multimodal imaging spectrometer of the high-resolution special aerial system, with a spectral range of 400–1000 nm [63]. This dataset consists of 250 bands, with an image size of 3750 × 1580 pixels, and has been labeled with 20 categories.

The KSC dataset was also collected by AVIRIS with wavelengths ranging from 400 to 2500 nm [62]. This image has a spatial dimension of 512 × 614 pixels and 176 spectral bands after removing low signal-to-noise ratio bands. Also, the KSC dataset consists of 5202 samples of 13 upland and wetland categories.

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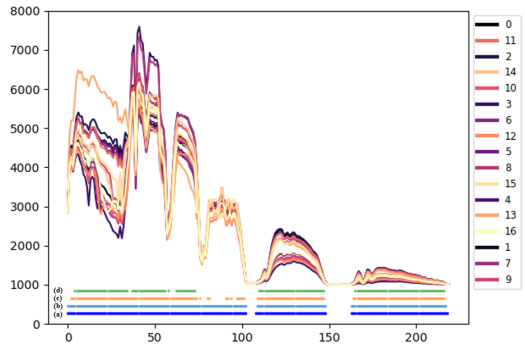


Fig. 8. Band exclusion results for the IP dataset with different methods compared with the related spectra: (a) traditional manual selection (dark blue dot lines), (b) our method with 90% (light blue dot lines), (c) our method with 80% (orange dot lines), and (d) our method with 70% reserved bands (green dot lines).

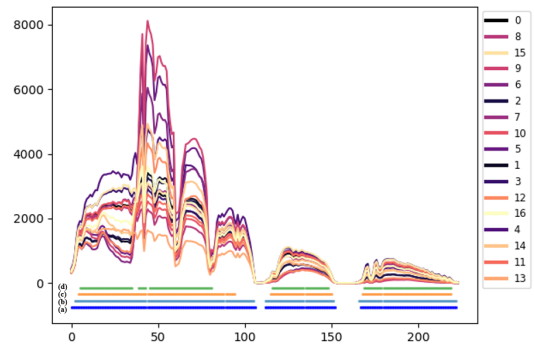


Fig. 9. Band exclusion results for the SA dataset with different methods compared with the related spectra: (a) traditional manual selection (dark blue dot lines), (b) our method with 90% (light blue dot lines), (c) our method with 80% (orange dot lines), and (d) our method with 70% reserved bands (green dot lines).

In each dataset, 10% of the sample data are randomly selected for the training and the rest for the testing, as the benchmark classification method in [64].

B. Experimental Results on Band Selection

Here, the results of band exclusion for the IP dataset [62] are visually exhibited in Fig. 8, where the reserved bands are shown as a dot line below the related spectra; the result of traditional manual selection is shown with dark blue dot lines, and the results from our method of 90%, 80%, and 70% reserved bands are shown with light blue, orange, and green colored dot lines, respectively.

First, we can see from the dot lines in Fig. 8 that the bands in the range of 104–108, bands 150–163 and 220, which are affected by water vapor, are generally removed from the IP dataset.

Second, the traditional reserved bands shown with dark blue dot lines and the bands reserved by our proposed method marked with other colored dot lines are basically consistent. It can be seen from Fig. 8 that the traditionally removed bands affected by water vapor and Rayleigh scattering are near the 108th and 158th bands. However, compared with the traditional manual selection, whose excluded bands are mainly near the range of 108 and 158, i.e., the two blank intervals, our proposed method can automatically remove the bands near these two ranges, that is, the bands excluded by our proposed method can completely cover the bands removed by the traditional manual selection shown with the dark blue dot lines. Specifically, the bands excluded by our proposed method are in the range of 75–100, corresponding to the wavelength range of 370–409 nm, which belongs to the blue band and may be weakened by the Rayleigh scattering. The wavelength range of bands 75–100 removed by our proposed method is in the range of 1069–1284 nm, which belongs to the near-infrared-to-short-wave-infrared band and may be weakened by water vapor.

More importantly, one of another band exclusion results for the SA hyperspectral image dataset [62] is shown in Fig. 9, and we can see from it that the removed weakened bands by our proposed method are different for the different hyperspectral image datasets with different time phases, which indicates that maybe our proposed method has some kinds of correlation with the imaging conditions, such as time, region, or the land cover types.

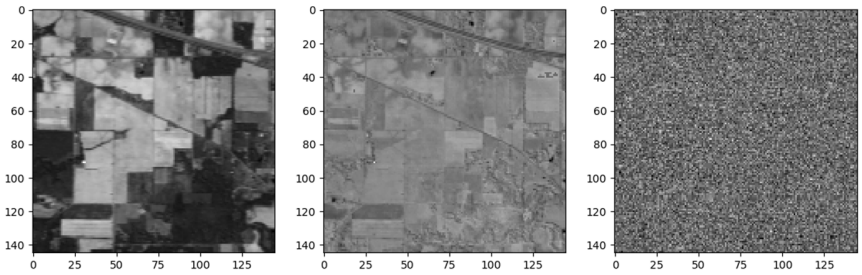
In order to intuitively observe the practical significance of our proposed method, the individual grayscale images corresponding to the bands with the maximum and minimum values of band weaken index in the IP dataset are shown in Fig. 10.

It can be seen from Fig. 10 that the band with large band weaken index corresponds to a grayscale image with strong contrast, which indicates that there are more high-frequency components in the power spectral density of the grayscale image that can better describe the changes within it, such as the edges. The band with a low value of the band weaken index does correspond to the weakened band, i.e., a grayscale image with weakened or flattened surface reflectance among the adjacent pixels, which is difficult to reflect the difference between different categories and has less significant value for the subsequent classification task. Therefore, after removing the weakened bands, specific bands that may be affected by the water vapor or Rayleigh scattering can be removed, and the remaining bands may be more suitable for the subsequent classification or other application tasks.

C. Experimental Results on Classification

Here, experiments are first conducted on the IP dataset, and the attention-based adaptive spectral–spatial kernel ResNet (A2S2K) method [64] is selected as the benchmark method of this hyperspectral image classification; their default parameters and structures are utilized to make a fair comparison. Overall accuracy (OA), average accuracy (AA), and Kappa coefficient are selected as the evaluation indexes of HSI classification, in which the OA is calculated as the total number of correctly classified pixels divided by the total number of pixels. The AA is calculated as the sum of the accuracy of each class divided by the number of classes. Also, the Kappa coefficient is an index for consistency test, to measure whether the predicted results of the model are consistent with the actual classification results. In this experiment, the threshold of spatial autocorrelation is set to reserve 80% of the total bands, i.e., here, in this experiment, we set $\beta_2 = 0.2 \times (\gamma_{\max} - \gamma_{\min})$. Also, the

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(a) (b) (c)

Fig. 10. Grayscale images of the bands corresponding to the maximum and minimum values and NaN of (a) band corresponds to max γ, (b) band corresponds to min γ, (c) band corresponds to NaN γ, the band weaken index in the IP dataset.

TABLE III
VALIDATION RESULTS ON THE IP DATASET

No.	Category	Original 220 bands	Traditional 200 bands	Our method with 176 (80%) bands
1	Alfalfa	**100.0**	97.56	99.19
2	Corn-notill	98.16	98.62	**99.23**
3	Corn-mintill	98.20	98.58	**98.82**
4	Corn	98.44	98.29	**98.30**
5	Grass-pasture	99.20	**99.02**	98.86
6	Grass-trees	98.88	98.71	**99.25**
7	Grass-pasture-mowed	90.63	93.10	**95.39**
8	Hay-windrowed	98.65	98.83	**100.0**
9	Oats	88.57	74.26	**92.77**
10	Soybean-notill	93.37	98.21	**98.29**
11	Soybean-mintill	98.21	**99.09**	99.05
12	Soybean-clean	96.61	98.37	**98.60**
13	Wheat	**100.0**	99.80	99.59
14	Woods	99.00	99.22	**99.37**
15	Buildings-Grass-Trees-Drives	97.88	97.86	**98.62**
16	Stone-Steel-Towers	**97.08**	95.93	96.42
OA		97.79	98.66	**98.97**
AA		97.06	96.59	**98.23**
Kappa		97.49	98.48	**98.82**

TABLE IV
VALIDATION RESULTS ON THE SA DATASET

No.	Category	Original 224 bands	Traditional 204 bands	Our method with 179 (80%) bands
1	Brocoli_green_weeds_1	99.97	**100.0**	**100.0**
2	Brocoli_green_weeds_2	**100.0**	**100.0**	**100.0**
3	Fallow	**100.0**	**100.0**	**100.0**
4	Fallow_rough_plow	**99.84**	99.76	99.80
5	Fallow_smooth	99.81	99.87	**99.94**
6	Stubble	99.98	99.99	**100.0**
7	Celery	99.90	**100.0**	**100.0**
8	Grapes_untrained	99.89	99.71	**99.98**
9	Soil_vinyard_develop	99.96	99.92	**99.99**
10	Corn_senesced_green_weeds	99.90	99.80	**99.93**
11	Lettuce_romaine_4wk	**100.0**	**100.0**	**100.0**
12	Lettuce_romaine_5wk	**100.0**	99.98	**100.0**
13	Lettuce_romaine_6wk	99.97	**100.0**	**100.0**
14	Lettuce_romaine_7wk	**99.82**	99.62	99.65
15	Vinyard_untrained	99.52	99.73	**99.92**
16	Vinyard_vertical_trellis	99.91	99.93	**100.0**
OA		99.87	99.86	**99.96**
AA		99.91	99.89	**99.95**
Kappa		99.86	99.84	**99.96**

accuracy of using all the original 220 bands, the traditional manual selected 200 bands, and the 176 (80%) bands from our proposed method are compared.

The experimental results are shown in Table III, with the best-performing results highlighted in bold. From Table III, it can be seen that OA, AA, and Kappa indexes are all improved on the IP dataset by using our proposed method. In particular, the improvement of the categories with lower classification accuracy by using the original bands is very obvious, e.g., category 7 has been improved by 4.66%, category 9 has been improved by 4.20%, and category 10 has improved by 4.92%.

Furthermore, the spectra of one specific category in the IP dataset before and after band exclusion are also given in the upper part of Fig. 8. It can be observed from Fig. 8 that, for the IP dataset, the largest changes in the spectra appeared is in the range of band 75–100, and they are from the near-infrared to short-wave infrared ranges, which are usually be applied to evaluate the vegetation coverage and biomass. However, the strong absorption of water vapor in these bands generally weakens the surface reflectance and affects the classification accuracy. As shown in Table III, after these bands are excluded, the influence of water vapor can be removed, and the accuracy of classification for the nongreen plants is increased, but a decrease in accuracy can also be found for category 1, due to the spectral response oscillations caused by the interaction of green vegetation and water vapor.

We also conducted a validation experiment on the SA dataset shown in Table IV. From Table IV, it can be seen that by using our proposed method, OA, AA, and Kappa indexes are all improved. Since the benchmark accuracy of the original SA dataset is already greater than 99.8%, the improvement on the SA dataset is smaller than that on the IP dataset. However, these experiments have verified that our proposed method

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TABLE V
VALIDATION RESULTS ON THE XA DATASET

Category	Original 256 bands	Our method with 204 (80%) bands
1	95.43	**96.87**
2	**98.89**	98.48
3	95.97	**97.41**
4	99.75	**99.91**
5	96.22	**97.63**
6	97.70	**98.35**
7	99.56	**99.79**
8	97.61	**99.60**
9	98.55	**99.56**
10	99.56	**99.87**
11	**96.67**	84.87
12	91.41	94.49
13	97.30	**97.59**
14	**91.51**	91.29
15	90.95	95.05
16	86.37	**86.56**
17	74.14	**84.20**
18	96.65	**97.39**
19	96.61	**97.34**
20	96.88	**98.95**
OA	97.13	**97.90**
AA	94.89	**95.76**
Kappa	96.67	**97.56**

applies to different hyperspectral image datasets with different time phases, which indicates that our proposed method has strong universality.

Furthermore, we also verify our proposed method on the XA dataset. To keep consistent, the A2S2K method [64] is also selected as the benchmark method, 80% of the bands are selected by our proposed method, and the results are shown in Table V. On the XA dataset, by using our proposed method, OA, AA, and Kappa indexes are all improved obviously, and the promotion on the categories with lower classification accuracy in the original dataset is also remarkable, e.g., category 17 has improved from 74.14 to 84.20 by 10.06%, category 15 has improved from 90.95 to 95.05 by 4.10%, and category 12 has improved from 91.41 to 94.49 by 3.08%. We should also mention that this experiment verified the effectiveness and universality of our proposed method for different platforms. It also should be highlighted that Fig. 11 gives the band exclusion results for the XA dataset compared with the related spectra, and the removed bands of XA dataset are quite different from that of IP dataset and SA dataset shown in Figs. 8 and 9, respectively, which indicates that our proposed method has strong universality.

In addition, we conducted validation experiments on the KSC dataset as well, in which the A2S2K method [64] is also selected as the benchmark classification method. Due to the original KSC dataset before removing water absorbing bands not being published, a traditional band selection with 176 bands is adopted in this experiment. Thus, our proposed method is equivalent to the secondary band exclusion and here also 80% of the bands are reserved. The experimental results for the KSC dataset are shown in Table VI, with

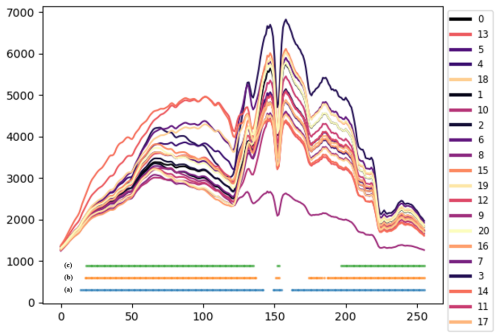


Fig. 11. Band exclusion results for XA dataset with different percentages of the reserved bands: (a) 90% (light blue), (b) 80% (orange), and (c) 70% reserved bands (green).

TABLE VI
VALIDATION RESULTS ON THE KSC DATASET

Category	Original 176 bands	Our method with 140 (80%) bands
1	**99.95**	99.92
2	98.68	**98.84**
3	98.72	**99.77**
4	94.27	**99.19**
5	94.46	**98.52**
6	99.82	**99.91**
7	99.61	**100.0**
8	**100.0**	99.86
9	**100.0**	**100.0**
10	**100.0**	**100.0**
11	**100.0**	99.95
12	**100.0**	**100.0**
13	**100.0**	**100.0**
OA	99.34	**99.81**
AA	98.88	**99.69**
Kappa	99.27	**99.79**

the best-performing results highlighted in bold. In Table VI, although there are some data fluctuations, the OA, AA, and Kappa indexes are all improved remarkably, and some categories, such as categories 4 and 5, have been improved even by 4.92% and 4.06%, respectively. This indicates that our proposed method can even serve as a secondary band exclusion method, to further remove the affected band after the traditional manual selection, which makes this proposed method widely applicable.

D. Improvement to the Other Classification Methods

In this section, in addition to the aforementioned A2S2K method, the improvement of our proposed method to the other benchmark methods of hyperspectral image classification is also verified. Here, the most basic PyResNet method [68] and the most popular SSRN method [66] are selected for this experiment, and the default parameters and structures suggested in their literature are utilized to make a fair comparison. The IP dataset is also utilized in this experiment, and the experimental results can be seen in Table VII, with the best-performing results highlighted in bold.

From Table VII, it can be seen that our proposed method is able to boost different classification methods for hyperspectral

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TABLE VII
IMPROVEMENT TO THE OTHER DIFFERENT HSI CLASSIFICATION METHODS

Category	PyResNet		SSRN		A2S2K	
	Original 220 bands	Our method	Original 220 bands	Our method	Original 220 bands	Our method
1	89.29	99.00	57.67	89.98	100.0	99.19
2	87.45	**92.12**	97.76	**99.18**	98.16	**99.23**
3	83.56	**89.34**	97.93	**98.14**	98.20	**98.82**
4	**96.48**	93.97	98.18	**98.88**	**98.44**	98.30
5	99.41	98.05	96.88	97.66	99.20	98.86
6	94.08	96.47	98.88	99.14	98.88	99.25
7	100.0	95.24	50.00	65.79	90.63	95.39
8	91.46	94.83	95.63	99.62	98.65	100.0
9	50.00	**90.00**	25.00	**99.02**	88.57	**92.77**
10	89.41	90.04	96.11	98.57	93.37	98.29
11	89.16	93.55	98.95	99.36	98.21	99.05
12	92.00	92.29	98.49	98.18	96.61	98.60
13	95.75	98.24	100.0	99.13	100.0	99.59
14	96.68	**97.45**	98.81	**99.37**	99.00	**99.37**
15	87.56	95.85	99.34	99.06	97.88	98.62
16	**97.31**	91.85	94.16	**95.99**	97.08	96.42
OA	90.04	**93.62**	97.99	**98.88**	97.79	**98.97**
AA	89.97	**94.27**	84.61	**96.07**	97.06	**98.23**
Kappa	88.61	**92.72**	97.71	**98.73**	97.49	**98.82**

TABLE VIII
COMPARISON WITH THE OTHER DIFFERENT BAND SELECTION METHODS

Category	IBRA-GSS	PLS-DA	SR-SSIM	OCF	FNGBS	WBE
1	39.49	25.00	**69.05**	50.00	37.92	57.30
2	93.23	89.07	94.03	92.38	**93.63**	93.48
3	93.28	**95.68**	93.67	93.69	90.75	93.40
4	92.89	90.89	91.13	93.52	87.47	**94.35**
5	93.70	89.97	86.68	89.41	92.62	95.06
6	90.01	93.41	91.51	95.17	96.51	**98.04**
7	40.50	25.00	56.41	22.50	23.44	**58.75**
8	92.95	94.16	94.22	95.28	97.17	94.78
9	36.11	0.00	51.14	31.88	41.67	48.98
10	89.82	91.17	91.16	89.85	93.20	94.64
11	88.46	94.97	94.42	94.29	94.87	95.20
12	91.69	84.78	89.68	93.36	86.75	93.09
13	96.90	94.74	94.72	92.25	96.42	96.48
14	89.96	95.56	**98.22**	96.48	97.85	97.62
15	86.61	87.31	91.16	90.48	91.06	97.10
16	90.75	94.43	95.69	**98.68**	92.47	93.55
OA	89.99	92.32	92.98	93.22	93.68	**95.10**
AA	81.65	77.88	86.43	82.45	82.11	**87.61**
Kappa	88.51	91.23	92.00	92.26	92.79	**94.41**

images remarkably. Even for the most basic PyResNet method, the improvement is remarkable, e.g., OA has improved from 90.04 to 93.62 by 3.58%, AA has improved from 89.97 to 94.27 by 4.30%, and Kappa has improved from 88.61 to 92.72 by 4.11%.

E. Comparison With Other Band Selection Methods

In this section, to verify the superiority of our proposed method, different popular band selection methods are compared on the IP dataset, including PLS-DA [21], SR-SSIM [34], OCF [32], FNGBS [25], and IBRA-GSS [67] methods. All these compared method have been briefly introduced in Section I, and their parameter settings keep consistent with that in IBRA-GSS [67]. According to IBRA-GSS's suggestion that using more than five bands in training would yield performance (OA $>$ 98%) and is not conducive to compare accuracy [67], in this experiment, only five bands are selected from the original 220 spectral

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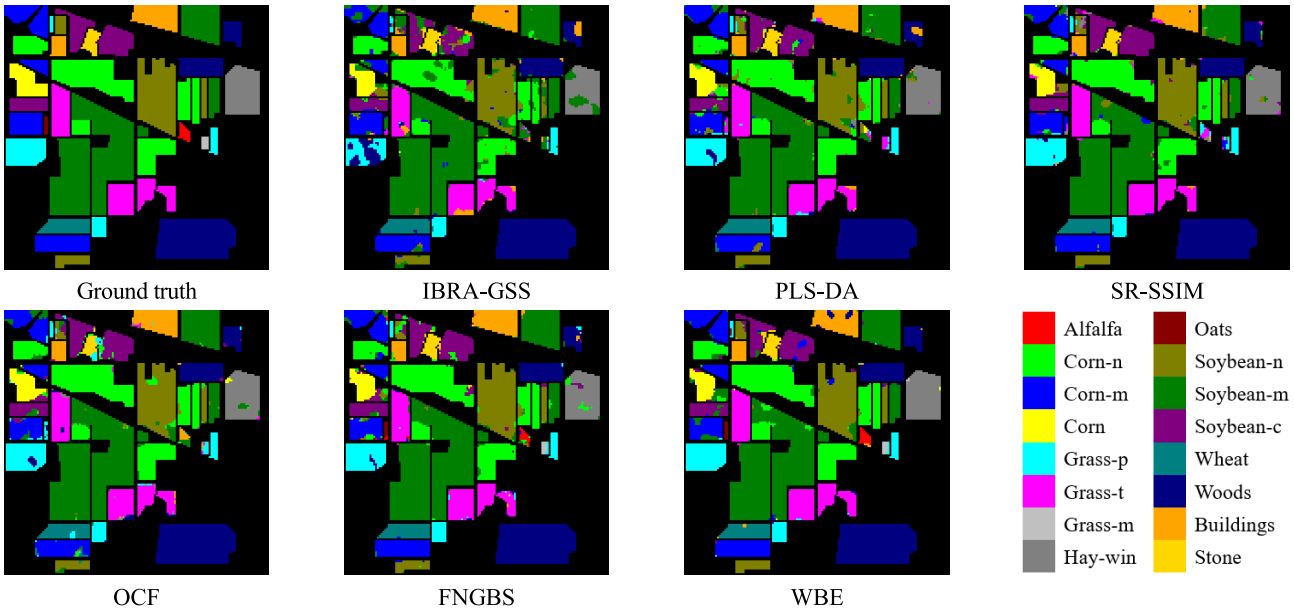


Fig. 12. Comparison with the other different band selection methods through HSI classification on the IP dataset.

TABLE IX
RELATIONSHIP BETWEEN THE ACCURACY OF CLASSIFICATION AND THE PERCENTAGE OF RESERVED BANDS ON THE IP DATASET

Category	30% reserved bands	40% reserved bands	50% reserved bands	60% reserved bands	70% reserved bands	80% reserved bands	90% reserved bands	Original 220 bands
1	97.46	99.19	**100.0**	**100.0**	97.46	99.19	97.33	**100.0**
2	97.54	98.46	98.78	98.33	98.41	99.23	98.55	98.16
3	97.80	98.64	98.59	98.05	98.28	**98.82**	97.56	98.20
4	98.58	98.09	98.37	98.66	98.54	98.30	**99.04**	98.44
5	97.89	98.99	98.53	98.51	98.52	98.86	98.53	99.20
6	98.58	98.66	**99.38**	98.86	99.02	99.25	98.74	98.88
7	96.81	94.95	97.53	97.02	**98.07**	95.39	93.94	90.63
8	99.16	99.70	99.62	99.96	99.70	**100.0**	99.85	98.65
9	84.69	89.70	90.63	91.53	92.40	**92.77**	81.91	88.57
10	96.87	98.13	97.73	98.00	97.91	98.29	98.28	93.37
11	98.11	98.22	98.17	98.87	99.08	99.05	99.07	98.21
12	98.20	98.42	**98.95**	97.92	98.35	98.60	97.41	96.61
13	99.69	99.10	99.09	99.01	100.0	99.59	99.51	100.0
14	99.22	99.53	99.27	99.48	99.39	99.37	99.51	99.00
15	98.46	98.46	99.35	99.46	99.14	98.62	99.10	97.88
16	**97.66**	97.40	95.71	97.44	96.94	96.42	96.44	97.08
OA	98.11	98.57	98.63	98.69	98.76	**98.97**	98.66	97.79
AA	97.29	97.85	98.10	98.19	98.20	**98.23**	97.17	97.06
Kappa	97.84	98.37	98.44	98.50	98.59	**98.82**	98.48	97.49

bands. In addition, A2S2K [64] is also selected as the benchmark classification method for the hyperspectral images, and the experimental results are shown in Table VIII, with the best-performing results highlighted in bold. Besides, comparison results with the other different band selection methods through HSI classification are shown in Fig. 12.

From Table VIII, it is evident that our proposed method outperforms the other competitive band selection methods. Especially, compared with the other method, the accuracy of category 7 is improved by 2.34%–36.25%; more importantly, the OA index is improved by 1.43%–5.12%, and the Kappa index is improved by 1.62%–5.90%, which validated the superiority of our proposed method.

F. Discussion on Parameter Setting

The major parameter setting concerning with our proposed method is the threshold of percentage. Since the improved accuracy on the IP dataset is remarkable, the experiment about

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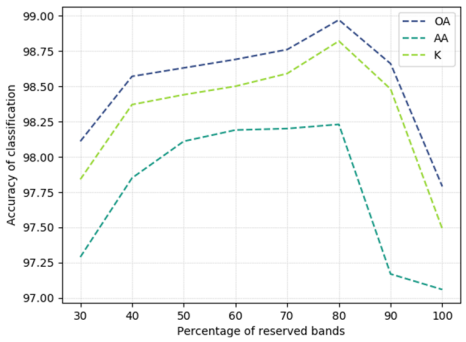


Fig. 13. Relationship between the accuracy of classification and the percentage of reserved bands on the IP dataset.

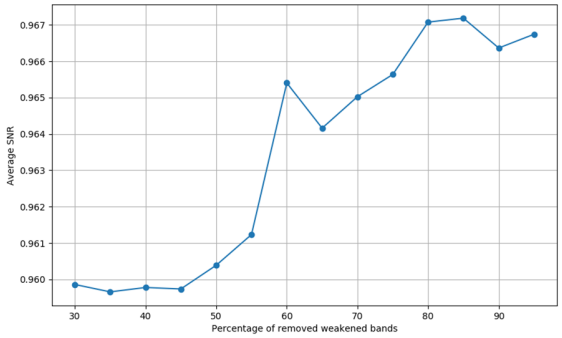


Fig. 14. Relationship between SNR and the percentage of the removed weakened bands on the IP dataset.

how to set the threshold of percentage will be conducted on it. The experimental results with different percentages of the reserved bands are given in Table IX and Fig. 13.

From Table IX, we can see that the accuracy of hyperspectral image classification can be further improved by our proposed method with the increasing percentage of reserved bands. From the relationship between the accuracy of classification and the percentage of reserved bands shown in Fig. 13, we can also see that a maximum accuracy can be reached when the percentage of the reserved bands is around 80%. That is why in the above experiments, we set the threshold of percentage to 80%. To the best of our knowledge, this is the first band selection method that can improve the HSI classification accuracy. In addition, the SNR of data in the IP dataset basically (but not strictly) increases with the increase of the removed weakened bands, as in the following Fig. 14, which has also validated the effectiveness of our proposed method.

IV. Conclusion

To remove the specific bands that are easily weakened by water vapor or Rayleigh scattering in the hyperspectral image classification, an evaluation-based weaken band exclusion method is proposed in this article. In detail, an evaluation model to describe how water vapor and Rayleigh scattering affect the surface reflectance is constructed, by using the statistic relationship between the radiative transfer model and the band weaken index of spectra among the adjacent pixels. Then, with a simulation experiment, it has been shown that water vapor and Rayleigh scattering can really weaken the discriminative information of specific bands, and the band weaken index can serve as an appropriate index to evaluate the weakening degree of those bands. On this basis, the overall framework of evaluation-based weaken band exclusion method is given. The effectiveness and the universality of our proposed method have been verified and compared on four representative hyperspectral image classification tasks.

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